



Generating a sequence using a position-to-term rule

1 A term-to term rule describes how to calculate the next term from.... Tick **1** correct answer

☐ the constant additive rule

☐ the multiplier

☒ the previous term

☐ the term number

2 True or false. In the sequence 0, 1, 2, 3, 4, ... the 2nd term is 3. Tick **1** correct answer

☐ True- the third number is '2'

☒ False- The second number is not '3'

☐ There is no way of knowing without the term-to-term rule

3 If the 7 times table start with $1 \times 7 = 7$ what is the 100th number in the 7 times table?
Fill in the blank

700



4 Here are the first 3 terms in a pictorial sequence. Which of these rules could refer to the number of lines in each term? Tick 2 correct answers



☐ add 2 to the previous term to get the next term

☒ add 3 to the previous term to get the next term

☐ add 5 to the previous term to get the next term

☒ there are 3 lines per section plus 2 for the base.

☐ there are 5 lines per section

5 Which of these is an advantage for using a term-to-term rule Tick 2 correct answers

☒ They are easy to use to find the next term in a sequence

☐ They are easy to use to find any term in a sequence

☒ They work for all sequences where there is a rule between successive terms

☐ They are unique, the same rule will always produce the same sequence.

☐ There is only one term-to-term rule that will work if you know the first 3 term.



6 Match the sequences with the term-to-term rules. Write the correct letter in each box

a	-2, -6, -18, -54, ...
b	-2, -6, -10, -14, ...
c	-2, -6, -14, -30, ...
d	-2, -6, -12, -20, ...
e	-2, -6, -14, -26, ...
f	-2, -6, -8, -8, ...

c	multiply the previous term by 2 then subtract 2
b	add - 4 to the previous term
a	multiply the previous term by 3
f	add - 4, add - 2, add 0, ...
e	add - 4, add - 8, add - 12, ...
d	add - 4, add - 6, add - 8, ...

